Cascade-Transform-Based Tensor Nuclear Norm for Hyperspectral Image Super-Resolution

Honghui Xu, *Member, IEEE*, Chuangjie Fang, Yilin Ge, Yubin Gu, and Jianwei Zheng, *Member, IEEE*

***Abstract*— Recent advancements in tensor nuclear norm (TNN) have led to promising solutions for hyperspectral image super-resolution (HSISR), which produces enriched outputs by fusing low-resolution hyperspectral images (LRHSIs) with high-resolution multispectral images (HRMSIs). However, current TNN, mainly reliant on the discrete Fourier transform (DFT), still suffers from mirroring boundary effects and singleton domain limitation. As a relief, we propose cascade-transform-based tensor nuclear norm (CTNN) with two variants for HSISR, featuring new definitions and algebraic structures for tensor product and TNN operations. The first variant processes tubal elements derived from DFT as inputs in the discrete cosine transform (DCT) domain, allowing for more nuanced feature extraction. The second learns adaptive matrices from the data in each iteration update and links them with a fixed DFT matrix to dynamically update the transform domain, preventing rank estimation bias. Furthermore, the nonconvex form of the proposed CTNN is applied to three modes of each spectral subspace similarity cube, termed log-sum-based full-scale CTNN (LFCTNN), capturing the global low-rank structure of LRHSI and the nonlocal similarities present in HRMSI. Experimental evaluations on various remote sensing datasets indicate that our approach exceeds existing state-of-the-art methods. The code is available at https://github.com/xuhonghui96/LFCTNN.**

***Index Terms*— Discrete Fourier transform (DFT), hyperspectral super-resolution, image fusion, self-adaptive, tensor nuclear norm (TNN).**

I. Introduction

HYPERSPECTRAL imaging (HSI) is an advanced modality that captures images across a multitude of narrow spectral bands, allowing for precise material identification based on distinct reflectance characteristics at various wavelengths [1], [2], [3], [4]. This capability has revolutionized various image analysis applications, including target recognition [5], [6], change detection [7], [8], and classification [9], [10]. Despite the significant potential, hyperspectral cameras are fundamentally constrained by a tradeoff between spectral and spatial resolution due to limitations in incident energy within optical remote sensing systems [11]. As a result, achieving high spectral resolution often necessitates a reduction in spatial resolution. Conversely, MSI offers higher spatial resolution but at the cost of spectral resolution. To counteract the dilemma, the most prevalent approach attempts to fuse spatial details from a high-resolution multispectral image (HRMSI) with spectral data from a low-resolution hyperspectral image (LRHSI) of the same scene, thereby generating a high-resolution HSI. This fusion technique, which can be seen as hyperspectral image super-resolution (HSISR), not only leverages the detailed spatial information from HRMSI and the comprehensive spectral data from LRHSI but also provides a cost-effective and practical solution to the intrinsic limitations of HSI sensors.

Recent HSISR methods can be principally categorized into two families: deep learning (DL)-based and model-based approaches. Within the DL framework, convolutional neural networks (CNNs) [12], [13], [14] and Transformers [15], [16], [17], [18] serve as foundational models. CNNs benefit from weight sharing and translation invariance, while Transformers employ global self-attention mechanisms, both contributing to high-performance outcomes. Dong et al. [15] unfolded the iterative HSI-SR algorithm into a model-guided deep convolutional network (MoG-DCN) and maintained the spatial–spectral consistency. In addition, Ma et al. [11] unveiled DCTransformer, a novel proposition that leverages cross-attention to mutually infuse information across different modalities, learning HSI priors effectively. Overall, DL-based methods are known for impressive fusion performance, yet evaluating network parameters requires extensive training samples, making the process highly complex and cumbersome. Besides, the generalizability of most current treatments is often limited by the specific model structure and the prior distribution of the training data. In addition, the black-box nature of DL also leaves poor interpretability.

Model-based [19], [20], [21], [22], [23], [24] approaches have garnered increasing interest due to their solid theoretical foundations. These methods typically formulate two key components—spatial–spectral fidelity and prior property constraints—followed by specific optimization algorithms for model updating. Various priors have been explored in recent years, including spectral correlation [25], local spatial smoothness [26], and global low rankness [27], [28],

Received 8 October 2024; revised 10 November 2024; accepted 22 November 2024. Date of publication 27 November 2024; date of current version 13 December 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62276232 and Grant 62406285, in part by China Postdoctoral Science Foundation under Grant 2024M752863, in part by the Pioneer and Leading Goose Research and Development Program of Zhejiang under Grant 2023C01241, and in part by the Key Program of Natural Science Foundation of Zhejiang Province under Grant LZ24F030012. *(Corresponding author: Jianwei Zheng.)*

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This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2024.3508456, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2024.3508456

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to name just a few. For example, Ma et al. [29] introduced a graph-based manifold regularization technique into both spatial and spectral components, integrating multimodal low-rank constraints to maintain spatial–spectral coherence and low-rank structures. Similarly, Lu et al. [30] developed a fusion model that synergizes adaptive total variation regularization with a superpixel-based weighted nuclear norm, effectively capturing both smoothness and low-rank characteristics concurrently. Considering the global correlations among the high-dimensional spectral bands of HSI, low-rank tensor representations and tensor rank as the constraint to resolute the underlying data is a natural way. Recently, tensor nuclear norm (TNN) [31], [32], [33], [34], [35], [36], [37], utilizing tensor singular value decomposition (TSVD) and closed multiplicative operations, has shown significant efficacy in capturing spatial-shifting correlations. Within this framework, Bu et al. [2] employed the TNN principle to elucidate the transferable subspace relationship between HRHSI and HRMSI, constructing a nonlocal patch tensor derived from multitensor low-rank subspaces to better capture the complex underlying structure of HSIs. Peng et al. [38] utilized the TSVD framework to decompose HRHSI into two low-dimensional subspace tensors—spatial and spectral—along with a low-rank coefficient tensor. Sparse regularization was subsequently imposed to refine the computation of the coefficient tensor, facilitating efficient and accurate image fusion. Dian et al. [25] generalized the third-order TNN to accommodate arbitrary modes and applied it to coefficient patches that simultaneously represent spatial, spectral, and nonlocal features, resulting in remarkable fusion performance. The aforementioned methods are built upon the TSVD framework based on discrete Fourier transform (DFT). To more effectively capture the low-rank features of tensors and mitigate the sensitivity of DFT to nonsmooth variations along the third mode, researchers have explored alternative transformations, such as the discrete cosine transform (DCT) [39], semireversible framelet transform [40], and unitary transform [41]. However, these transformations are primarily designed for tasks such as color image processing or tensor completion and lack specialized adaptability for HSI, which are characterized by rich and complex spectral profiles that consistently exhibit significant spatial shift-related patterns, and the boundary mirror effect of DFT remains unresolved. In this scenario, fixed transformations based on DFT or other single matrix transformations within TNN may fail to exhibit sufficiently low-rank properties, thus diminishing the efficacy of fusion methods based on low-rank tensors.

To obtain precise estimations, we propose two types of cascade-transform-based TNN (CTNN) models for HSI-SR in a unified optimization framework and put forward new definitions and algebraic structures of the tensor product and TSVD operation, which is fundamentally anchored in DFT. In the first attempt, we substitute the DFT matrix in tensor SVD with a cascade transformation matrix comprising DFT and DCT, exploiting the mirror symmetry of the DCT to mitigate the boundary-blurring limitations inherent to DFT, while retaining the DFT's capacity for efficient high-frequency information capture. In the second attempt, we employ a computationally tractable self-learning cascaded TSVD within a rank minimization framework. This method integrates an adaptive learning matrix, updated iteratively, to enhance the DFT matrix, thus capturing the local structure of the underlying tensor subspace and reducing the rank estimation bias inherent to DFT. This method dynamically evolves, fostering self-learning properties and incorporating dual-layer prior information. In both types of cascaded transformations, the output from the first stage serves as the input for the second, redistributing the transformed data into tensor matrix slices aligned along the tubal direction. Each transformed matrix slice is subsequently subjected to singular value decomposition to elucidate its decomposition characteristics. Fig. 1 presents the Fourier magnitude spectrum in different transformation domains. It can be observed that DFT exhibits significant color variation, with prominent red and yellow regions, indicating a relatively complex transformed spectrum that is less conducive to low-rank exploration. In contrast, DCT shows a reduction in frequency components, resulting in a smoother appearance, which suggests a decrease in high-frequency interference. Moreover, the DFT-DCT (FC) and DFT-self-learning (FL) transformations exhibit more blue regions, demonstrating a sparser information distribution and a simpler data structure, which are more favorable for low-rank property exploration. Furthermore, to mitigate excessive penalization of singular values and reduce directional sensitivity, we introduce two log-sum-based forms of the proposed CTNN, applied across three tensor modes, which are termed the log-sum-based full-scale CTNN based on FC and FL cascade transformations (LFCTNN-FC/FL).

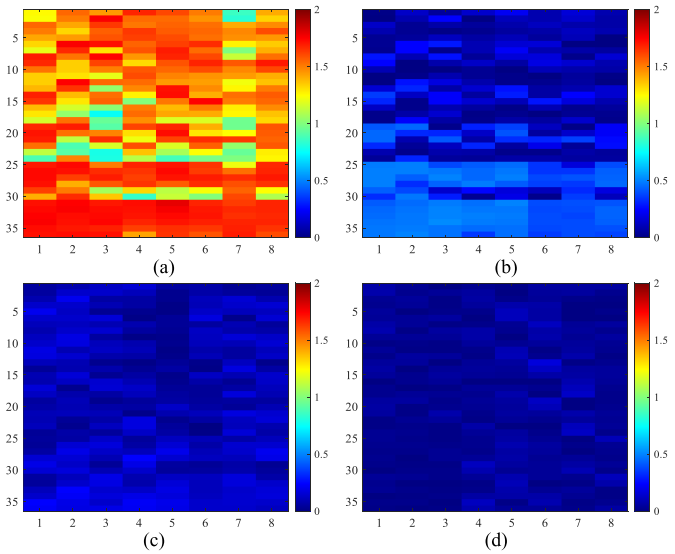


Fig. 1. Fourier magnitude spectrum of different transforms. (a) DFT, (b) DCT, (c) DFT-DCT, and (d) DFT-self-learning cascaded transform.

Based on the definitions of these two LFCTNN-FC/FL methods, we propose a novel CTNN and subspace-based HSISR method that can comprehensively explore multiple levels of information across three permutations of given tensors, supported by spectral low rankness and nonlocal similarities. Specifically, the spectral subspace coefficients and low-rank spectral bases are estimated from LRHSI via SVD, allowing for the partitioning and aggregation of full-band cubes of

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predetermined size. These spectral bases are then multiplied by the corresponding coefficients to reconstruct the target high-resolution spectra. Subsequently, to effectively capture the spatial and spectral similarities within the generated high-resolution HSI, a clustering approach is employed to classify similar blocks within the MSI. The classified blocks guide the organization of coefficients, which are compiled into a low-rank 3-D tensor encompassing spatial, spectral, and nonlocal dimensions. To preserve these intrinsic similarities during enhancement, the proposed LFCTNN-FC/FL is applied to all groups of distinct cubes derived from the subspace coefficients, thereby reframing the HSISR task as a low CTNN regularization optimization problem for the coefficients, which is addressed through iterative optimization techniques.

Our contribution can be summarized as follows.

1) We put forward a new definition of cascade-transform-based TSVD decomposition given a two-layer transformation matrix, from which algebraic operations associated with the 3-D tensor, such as cascade transform-oriented t-product, t-SVD, and low-rank approximation, can be intuitively derived.
2) By leveraging these newly defined constructs, we propose two types of log-sum cascade-transform-based TNN: LFCTNN-FC and LFCTNN-FL, both designed to induce minimum-rank tensors in the transformed domain. In addition, the new TNN models are applied to three permuted forms of the tensor constructed through clustering, aiming to simultaneously explore spatial low-rank, spectral low-rank, and nonlocal characteristics.
3) We develop an efficient alternating direction method of multipliers (ADMM) to solve the optimization of the proposed model. Numerical experiments substantiate that our proposal surpasses the majority of contemporary approaches in terms of fusion performance.

II. Notations and Preliminaries

In this section, we begin with a brief introduction to the notations and fundamental definitions presented in Table I. We then proceed to define the concepts of TNN and TSVD, along with the original formulation of HSISR.

A. TSVD and TNN

Definition 1 (t-Product [42]): For any two given tensors $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ and $\mathcal{Y} \in \mathbb{R}^{I_2 \times I_4 \times I_3}$, the t-product $\mathcal{Z}$ of $\mathcal{X}$ and $\mathcal{Y}$ is denoted as a tensor with size $I_1 \times I_4 \times I_3$, i.e.,

$$\mathcal{Z} = \mathcal{X} * \mathcal{Y} = \text{fold}(\text{bcirc}(\mathcal{X}) \cdot \text{unfold}(\mathcal{Y})) \tag{1}$$

where $\cdot$ is the matrix product.

In fact, the t-product is analogous to the matrix product with an additional circulant convolution operation; hence, (1) can also be represented in the Fourier domain as

$$\mathcal{Z} = \mathcal{X} * \mathcal{Y} \Longleftrightarrow \bar{\boldsymbol{Z}} = \bar{\boldsymbol{X}}\bar{\boldsymbol{Y}} \tag{2}$$

where the terms $\bar{\boldsymbol{X}}$, $\bar{\boldsymbol{Y}}$, and $\bar{\boldsymbol{Z}}$ are all block diagonal matrices whose frontal slices are diagonal blocks.

TABLE I
Notation Declarations

Notations	Explanations
$\mathbf{x}, \boldsymbol{X}, \mathcal{X}$	Vector, matrix, tensor
$\boldsymbol{X}_{(k)}$	The mode-k unfolding of $\mathcal{X}$
$\mathcal{X} \times_k \boldsymbol{U}$	the mode-k product: $\mathcal{X} \times_k \boldsymbol{U} \Leftrightarrow \boldsymbol{U} \cdot \boldsymbol{X}_{(k)}$
$\bar{\mathcal{X}} = \text{fft}(\mathcal{X}, [\,], 3)$	the DFT of $\boldsymbol{\mathcal{X}}$ along with mode-3
$\bar{\boldsymbol{X}}^{(i)}$	$\bar{\boldsymbol{X}}^{(i)} = \bar{\mathcal{X}}(:,:,i),\ i = 1, 2, \ldots, I_3$
$\text{bcirc}(\mathcal{X}) \in \mathbb{R}^{I_1 I_3 \times I_2 I_3}$	$\begin{bmatrix} \boldsymbol{X}^{(1)} & \boldsymbol{X}^{(I_3)} & \cdots & \boldsymbol{X}^{(2)} \\ \boldsymbol{X}^{(2)} & \boldsymbol{X}^{(1)} & \cdots & \boldsymbol{X}^{(3)} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{X}^{(I_3)} & \boldsymbol{X}^{(I_3-1)} & \cdots & \boldsymbol{X}^{(1)} \end{bmatrix}$
$\text{bdiag}(\bar{\mathcal{X}}) \in \mathbb{R}^{I_1 I_3 \times I_2 I_3}$	$\begin{bmatrix} \bar{\boldsymbol{X}}^{(1)} & & & \\ & \bar{\boldsymbol{X}}^{(2)} & & \\ & & \ddots & \\ & & & \bar{\boldsymbol{X}}^{(I_3)} \end{bmatrix}$,
$\text{unfold}(\mathcal{X}) \in \mathbb{R}^{I_1 I_3 \times I_2}$	$\text{unfold}(\mathcal{X}) = \left[\boldsymbol{X}^{(1)}, \boldsymbol{X}^{(2)}, \cdots, \boldsymbol{X}^{(I_3)}\right]^{\mathrm{T}}$
fold($\mathcal{X}$)	takes unfold($\mathcal{X}$) back to original tensor form
$\mathcal{X}^1 \in \mathbb{R}^{I_3 \times I_2 \times I_1}$	permute-1 operation of $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$
$\mathcal{X}^2 \in \mathbb{R}^{I_1 \times I_3 \times I_2}$	permute-2 operation of $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$
$\mathcal{X}^3 \in \mathbb{R}^{I_1 \times I_2 \times I_3}$	permute-3 operation of $\mathcal{X}$, $\mathcal{X}^3 = \mathcal{X}$
$\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$	inverse operation: ipermute $\left(\mathcal{X}^k, k\right)$

Definition 2 (t-SVD [42]): Any given tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ can be factorized as

$$\mathcal{X} = \mathcal{U} * \mathcal{S} * \mathcal{V}^{\mathrm{T}} \tag{3}$$

where $\mathcal{U} \in \mathbb{R}^{I_1 \times I_1 \times I_3}$ and $\mathcal{V} \in \mathbb{R}^{I_2 \times I_2 \times I_3}$ are orthonormal tensors and $\mathcal{S} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ is an f-diagonal tensor whose frontal slices are diagonal matrices.

Definition 3 (Tensor Nuclear Norm [43]): For any tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, TNN is denoted as the summation of singular values of each frontal slice of $\bar{X}$

$$\|\mathcal{X}\|_{\text{TNN}} = \frac{1}{I_3} \sum_{i=1}^{I_3} \|\bar{\boldsymbol{X}}^{(i)}\|_*. \tag{4}$$

B. Original Problem Formulation

In this study, our goal is to reconstruct the target HRHSI $\mathcal{X} \in \mathbb{R}^{W \times H \times S}$ by merging the obtained LRHSI $\mathcal{Y} \in \mathbb{R}^{w \times h \times S}$ and the corresponding HRMSI $\mathcal{Z} \in \mathbb{R}^{W \times H \times s}$ from the same scene. Here, W and H refer to the spatial dimensions of HRHSI and HRMSI, while $w < W$ and $h < H$ denote the spatial dimensions of LRHSI. The symbols S and s represent the spectral bands of HR-HSI and HR-MSI, respectively. The LRHS $\mathcal{Y}$ can be considered as a downsampled version of $\mathcal{X}$ concerning the spatial dimension, expressed as

$$\boldsymbol{Y}_{(3)} = \boldsymbol{X}_{(3)} \boldsymbol{B} \boldsymbol{S} \tag{5}$$

where $\boldsymbol{Y}_{(3)} \in \mathbb{R}^{S \times hw}$ and $\boldsymbol{X}_{(3)} \in \mathbb{R}^{S \times HW}$ are the third-mode unfolded matrices of $\mathcal{Y}$ and $\mathcal{X}$, respectively. The spatial downsampling matrix $\boldsymbol{S} \in \mathbb{R}^{HW \times hw}$ selects the corresponding spectral pixels. The block circulant matrix $\boldsymbol{B} \in \mathbb{R}^{HW \times HW}$, which is diagonalized using the fast Fourier transform (FFT), accounts for convolution blur and is represented by

$$\boldsymbol{B} = \boldsymbol{F} \boldsymbol{K} \boldsymbol{F}^{-1} \tag{6}$$

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where the diagonal matrix $\boldsymbol{K}$ contains the eigenvalues of $\boldsymbol{B}$, and $\boldsymbol{F}$ and $\boldsymbol{F}^{-1}$ denote the FFT and its inverse, respectively.

To acquire the HR-MSI $\mathcal{Z}$, multispectral imaging sensors capture the same scene, modeling the HR-MSI as a spectrally downsampled version of $\mathcal{X}$

$$\boldsymbol{Z}_{(3)} = \boldsymbol{R}\boldsymbol{X}_{(3)} \tag{7}$$

where $\boldsymbol{Z}_{(3)} \in \mathbb{R}^{s\times HW}$ is the unfolded matrix of $\mathcal{Z}$ along the third dimension and $\boldsymbol{R} \in \mathbb{R}^{s\times S}$ represents the spectral response matrix of the MSI sensor. Given that HSI typically exhibits strong correlations across various bands, the global spectral low-rank characteristic serves as a key prior in HSI super-resolution. Therefore, [44] use the subspace decomposition technology for HRHSI to reveal the spectral low-rank property, i.e., $\boldsymbol{X}_{(3)} = \boldsymbol{D}\boldsymbol{A}_{(3)}$, where $\boldsymbol{D} \in \mathbb{R}^{S\times L}$ is low-dimensional spectral basis and $\boldsymbol{A} \in \mathbb{R}^{L\times HW}$ is the reduced image. Thus, the spatial and spectral degradation models can be reformulated as

$$\boldsymbol{Y}_{(3)} = \boldsymbol{D}\boldsymbol{A}_{(3)}\boldsymbol{B}\boldsymbol{S}, \boldsymbol{Z}_{(3)} = \boldsymbol{R}\boldsymbol{D}\boldsymbol{A}_{(3)}. \tag{8}$$

The HSISR problem can be reformulated as an estimation problem for the coefficient matrix $\boldsymbol{A}_{(3)}$, expressed as

$$\min_{\boldsymbol{A}_{(3)}} \|\boldsymbol{Y}_{(3)} - \boldsymbol{D}\boldsymbol{A}_{(3)}\boldsymbol{B}\boldsymbol{S}\|_F^2 + \|\boldsymbol{Z}_{(3)} - \boldsymbol{R}\boldsymbol{D}\boldsymbol{A}_{(3)}\|_F^2 + \lambda R\big(\boldsymbol{A}_{(3)}\big) \tag{9}$$

where $R(\cdot)$ denotes the regularization term and λ is the trade-off parameter. Incorporating various forms of prior knowledge is essential to constrain the solution space, mitigating the inherent indeterminacy of the HSI-MSI fusion task, which is the core pursuit of our work.

III. Main Result

In this section, we first present the algebraic framework for the cascade-transform-based TSVD and its convex surrogate, CTNN. Rather than applying rank constraints directly to the target tensor, the proposed CTNN is applied to each similar 3-D tensor patch. Finally, we introduce an ADMM-based solver for model optimization. The entire process is given in Fig. 2.

A. From DFT to the Cascaded Transform

Recently, DFT-based TSVD has demonstrated exceptional performance and has been widely applied. However, the traditional DFT-based TNN is limited by mirror boundary conditions and inherent constraints, such as the static and one-sided nature of the transformation matrix, which may result in incomplete data representation. In this work, we propose two DFT transformation cascade strategies. The first strategy, DFT-DCT cascade-transform-based TSVD, leverages the ability of DCT to concentrate most energy into fewer coefficients and its reduced sensitivity to boundary effects. The second strategy, DFT-self-learning cascade-transform-based TSVD, employs an iterative adaptive approach to replace the static and cumbersome DFT.

Using the slicewise low-rank coding and DFT technique, the original TNN can be computed via the summation of the matrix nuclear norm of $(\mathcal{X} \times_3 \boldsymbol{F}_{I_3})$'s frontal slices, where $\boldsymbol{F}_{I_3}$ is the DFT matrix. That is, (4) can be rewritten as $\|\mathcal{X}\|_{\text{TNN}} = \sum_{i=1}^{I_3} \|(\mathcal{X}\times_3 \boldsymbol{F}_{I_3})^{(i)}\|_*$. By replacing the transformation matrix, we can obtain TNN in other domains. Next, we will give some necessary definitions and properties of cascaded transform TSVD (CTSVD) and TNN based on the proposed cascade matrices.

Definition 4 (DFT-DCT Cascaded Transform): Given the DFT matrix $\boldsymbol{F}_{I_3} \in \mathbb{R}^{I_3\times I_3}$ and the DCT matrix $\boldsymbol{D}_{I_3} \in \mathbb{R}^{I_3\times I_3}$, the linear transform based on the cascade matrix $\boldsymbol{L}_{I_3} = \boldsymbol{F}_{I_3} \times \boldsymbol{D}_{I_3}$ is defined as $C_1 : \mathbb{R}^{I_1\times I_2\times I_3} \to \mathbb{R}^{I_1\times I_2\times I_3}$, which gives $C_1(\mathcal{X})$ by performing a linear transform on $\mathcal{X}$ along the third dimension, i.e.,

$$C_1(\mathcal{X}) = \mathcal{X} \times_3 \boldsymbol{L}_{I_3} \tag{10}$$

where the linear transform C_1 is given by $\boldsymbol{L}_{I_3} \in \mathbb{R}^{I_3\times I_3}$ that is invertible. We also have the inverse mapping given by

$$C_1^{-1}(\mathcal{X}) = \mathcal{X} \times_3 \boldsymbol{L}_{I_3}^{-1}. \tag{11}$$

Definition 5 (DFT-Self-Learning Cascaded Transform): Given any tensor $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times I_3}$, the self-learning transform matrix is achieved by applying dynamic SVD to the Mode-3 unfolding of target tensor. Subsequently, in each iteration, SVD operations are performed to update the transformation matrices, enabling adaptive learning of accurate information from progressively filled images. The specific computational procedure is simply given as follows:

$$\big[\boldsymbol{U}_{I_3}^t, , \big] = \text{SVD}\big(\boldsymbol{X}_{(3)}^t\big) \tag{12}$$

where t is the iteration number. $\boldsymbol{U}_{I_3}^t \in \mathbb{R}^{I_3\times I_3}$ is the self-learning matrix at the tth iteration with $\boldsymbol{U}_{I_3}\boldsymbol{U}_{I_3}^{\text{T}} = \boldsymbol{I}_{I_3}$. After obtaining the self-learning matrix $\boldsymbol{U}_{I_3}$, the DFT-self-learning cascade transform is defined as $C_2 : \mathbb{R}^{I_1\times I_2\times I_3} \to \mathbb{R}^{I_1\times I_2\times I_3}$, which gives $C_2(\mathcal{X})$ by performing a linear transform on $\mathcal{X}$ along the third dimension, i.e.,

$$C_2(\mathcal{X}) = \mathcal{X} \times_3 \boldsymbol{S}_{I_3} \tag{13}$$

where $\boldsymbol{S}_{I_3}$ represents the product of $\boldsymbol{F}_{I_3}$ and $\boldsymbol{U}_{I_3}$. Similarly, the inverse form of C_2 is $C_2^{-1}(\mathcal{X}) = \mathcal{X} \times_3 \boldsymbol{S}_{I_3}^{-1}$.

Before investigating low-rank properties, we can assess the favorable conditions for low-rank exploration by examining the cumulative energy (CE) versus frequency index curve (highlighted by the green dashed box in Fig. 2). The CE distribution of the DFT is notably linear, with energy evenly distributed across the entire frequency range, indicating significant potential for low-rank exploration. In contrast, the DFT+DCT demonstrates a slightly steeper increase in CE within the frequency index range of 100–200, reflecting a higher concentration of energy in the lower frequency region. Notably, the DFT with self-learning transformation exhibits a rapid increase in CE at lower frequencies, suggesting an even greater concentration of energy in low-frequency components, which facilitates the exploration of low-rank properties.

Based on the above phenomenon, using cascade matrices[1] $\boldsymbol{L}_{I_3}$ and $\boldsymbol{S}_{I_3}$, we then give some necessary definitions and properties.

[1]We will denote these two matrices collectively as $\widetilde{\boldsymbol{C}}$ in the following content.

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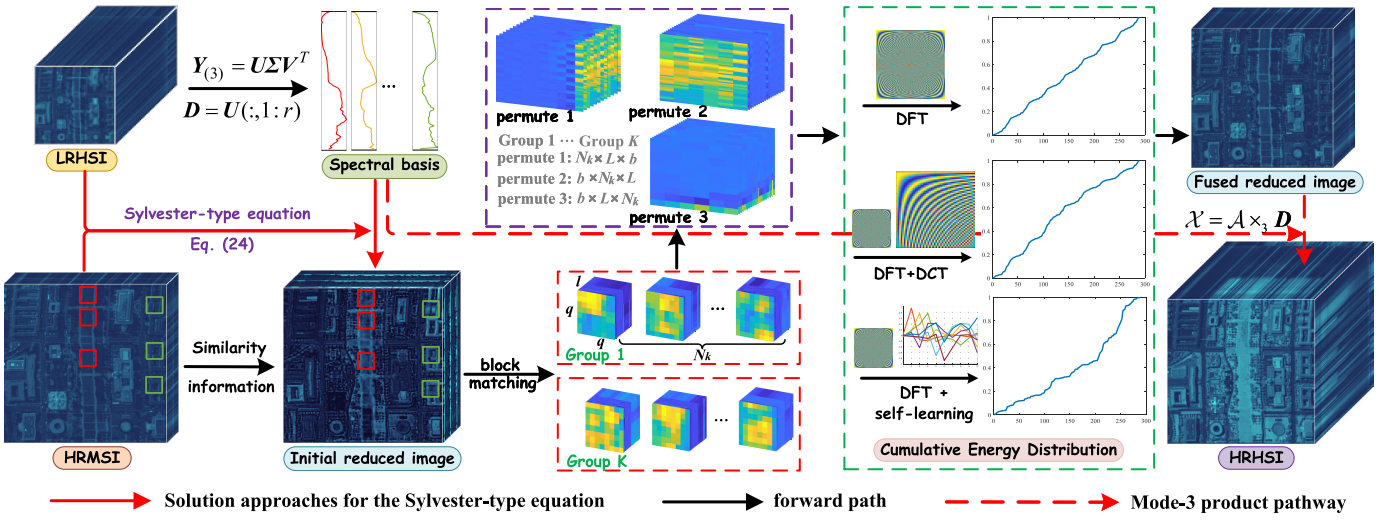


Fig. 2. Flowchart of the proposed CTNN method for HSISR.

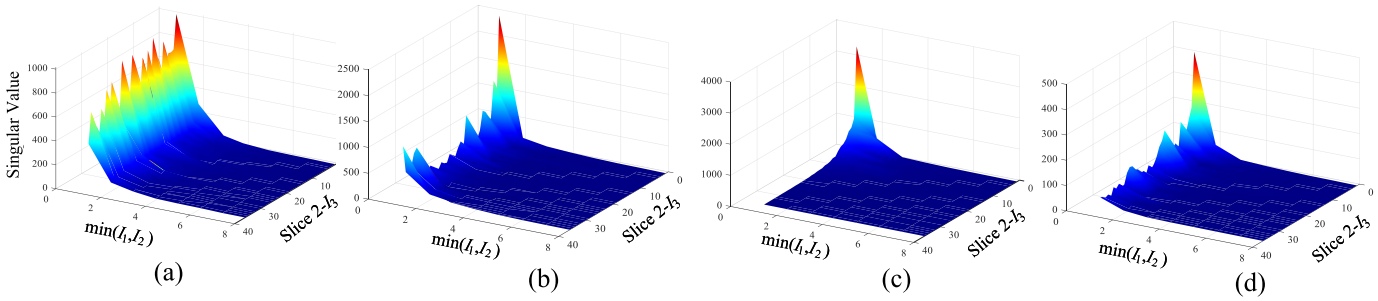


Fig. 3. Plot of singular values from using (a) DFT-DCT, (b) DFT-self-learning, (c) DFT, and (d) DCT.

Definition 6 (Cascaded Transform t-Product): Let C be the cascade transform, $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, and $\mathcal{Y} \in \mathbb{R}^{I_2 \times l \times I_3}$, and the t-product of $\mathcal{X}$ and $\mathcal{Y}$ under C, denoted as $\mathcal{Z} = \mathcal{X} *_C \mathcal{Y}$, is defined such that $C(\mathcal{Z}) = C(\mathcal{X}) \odot C(\mathcal{Y})$.

Definition 7 (Cascaded Transform Tensor Transpose): The tensor transpose of $\mathcal{X}$ with respect to C is the tensor $\mathcal{X}^{\mathrm{T}}$, satisfying $C(\mathcal{X}^{\mathrm{T}})^{(i)} = (C(\mathcal{X})^{(i)})^{\mathrm{T}}, i = 1, \ldots, I_3$.

Definition 8 (Cascaded Transform Identity Tensor): Let C be the cascade transform in (10) and (13), and $\mathcal{I} \in \mathbb{R}^{I \times I \times I_3}$ is called as identity tensor if it satisfies $\mathcal{I}(:,:,i) = \boldsymbol{I}$ for all face slices.

Definition 9 (Cascaded Transform Orthogonal Tensor): Under cascade transform matrix C, if $\mathcal{X}^{\mathrm{T}} *_C \mathcal{X} = \mathcal{X} *_C \mathcal{X}^{\mathrm{T}} = \mathcal{I}$, then the tensor $\mathcal{X}$ is the orthogonal tensor.

Definition 10 (f-Diagonal Tensor): A tensor is called f-diagonal if each of its frontal slices is a diagonal matrix.

Definition 11 (Cascaded Transform TSVD): Given $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ with cascade transform matrix C in (10) and (13), its CTSVD is defined as

$$\mathcal{X} = \mathcal{U} *_C \mathcal{S} *_C \mathcal{V}^{\mathrm{T}} \tag{14}$$

where $\mathcal{U}$ and $\mathcal{V}$ are orthogonal tensors and $\mathcal{S}$ is the f-diagonal tensor with respect to C.

Definition 12 (Cascaded Transform Tensor Tubal Rank): For any $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, let $\mathcal{S}$ be from the $\mathcal{X} = \mathcal{U} *_C \mathcal{S} *_C \mathcal{V}^{\mathrm{T}}$, and then, the CTSVD rank of $\mathcal{X}$ is defined as

$$\mathrm{rank}_{\mathrm{CTSVD}}(\mathcal{X}) := \sharp\{i, \mathcal{S}(i,i,:) \neq \mathbf{0}\} \tag{15}$$

where $\sharp$ denotes the cardinality of a set.

Definition 13 (Cascaded Transform Tensor Average Rank): Given $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ with cascade transform matrix $\widetilde{\boldsymbol{C}}$ in (10) and (13), the tubal rank of $\mathcal{X}$ is a vector $r \in \mathbb{R}^{I_3}$, whose ith element equals to the rank of the ith frontal slice of $C(\mathcal{X})$, i.e., $rank_a(\mathcal{X}) = (1/I_3) rank(\ddot{\boldsymbol{X}})$, where $\ddot{\boldsymbol{X}} = \mathrm{bdiag}(\mathcal{C}(\mathcal{X}))$.

To verify the effectiveness of CTSVD, Fig. 3 presents the low rankness of different transforms in the Pavia University (PU) dataset. As observed, a rapid decrease occurs in the singular values of the slices in the DFT-DCT cascade transform [see Fig. 3(a)], which verifies the approximately low rankness of these slices. The standalone DFT, DCT, and DFT-self-learning cascade transform demonstrate exceptionally strong singular value decay in the initial layers, yet their performance diminishes in the later bands, indicating a lack of sufficient low-rank characteristics across all bands. Notably, among the three transforms, the DFT-self-learning cascade transform exhibits the steepest rate of decay, suggesting that the proposed concatenation strategy is effective in enhancing the low-rank properties of the transform.

Since $\mathrm{rank}_{\mathrm{CTSVD}}(\mathcal{X})$ is nonconvex and even discrete, its convex relaxation, i.e., CTNN, can be deduced.

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Definition 14 (Tensor Nuclear Norm): Given tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ with the cascade transform matrix $\widetilde{\boldsymbol{C}}$, the CTNN is formulated as

$$\|\boldsymbol{\mathcal{X}}\|_{\text{CTNN}} = \frac{1}{I_3} \sum_{i=1}^{I_3} \|\left(\mathcal{X} \times_3 \widetilde{\boldsymbol{C}}\right)^{(i)}\|_*. \tag{16}$$

The following theorem shows that the CTNN of a tensor is the convex envelope of the entrywise summation of the cascade-transformed multirank over a unit ball of the tensor spectral norm.

Theorem 1: On the set $\{\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3} | \|C(\mathcal{X})\| \leq 1\}$, the convex envelope of the tensor average rank with respective to C is the tensor cascade transform nuclear norm $\|\mathcal{X}\|_{\text{CTNN}}$.

This underscores the effectiveness of CTNN in recovering tensors with low transformed tubal rank, as shown in the proof in the Supplementary Material.

B. Enhanced Cascade-Transform-Based TNN

Based on the observation model (9) and the proposed CTNN, the estimation of reduced image $\boldsymbol{A}_{(3)}$, where the rich image properties can be completely explored, is implemented by minimizing the energy function as follows:

$$\min_{\boldsymbol{A}_{(3)}} \|\boldsymbol{X}_{(3)} - \boldsymbol{D}\boldsymbol{A}_{(3)}\boldsymbol{B}\boldsymbol{S}\|_F^2 + \|\boldsymbol{Y}_{(3)} - \boldsymbol{R}\boldsymbol{D}\boldsymbol{A}_{(3)}\|_F^2 + \lambda \|\mathcal{A}\|_{\text{CTNN}}. \tag{17}$$

However, when applied to real-world scenarios with complex environments, relying solely on a single low-rank property tends to fall short of capturing intrinsic features. Therefore, we adopt a natural solution by introducing a nonlocal prior, which is highly effective for complex textured images, and propose a more comprehensive full-scale CTNN (FCTNN), applied to similar image patches. The MSI, with its preserved spatial structures, provides a foundation for exploring nonlocal spatial similarities. To facilitate this analysis, we segment the MSI into patches of size $(b)^{1/2} \times (b)^{1/2} \times s$ and subsequently organize these patches into K clusters, represented as $\boldsymbol{Z}^{(k)} = \{\mathcal{Z}^{(k,j)}\}_{j=1}^{N_k}$, where $k = 1, 2, \ldots, K$, with N_k indicating the number of patches within the kth cluster. The clustering is achieved through the K-means++ algorithm. The reduced image $\boldsymbol{A}_{(3)}$, which reflects the spatial characteristics of the HR-HSI, is constructed by mapping the self-similarities observed in the MSI onto these abundances. Using the established MSI cluster structure, we derive clusters of abundance patches, denoted as $\{\mathcal{A}^{(k,j)} \in \mathbb{R}^{(b)^{1/2} \times (b)^{1/2} \times L}\}_{j=1}^{N_k}$. These abundance patches, $\mathcal{A}^{(k,j)}$, share spatial coordinates with their MSI counterparts, $\mathcal{Z}^{(k,j)}$. The patches within each cluster $\mathcal{A}^{(k,j)}$ are assembled into a 3-D tensor $\mathcal{A}^k \in \mathbb{R}^{b \times L \times N_k}$, which encapsulates spatial, spectral, and nonlocal information. In this way, we estimate the reduced image by tackling the problem

$$\min_{\boldsymbol{A}_{(3)}} \|\boldsymbol{X}_{(3)} - \boldsymbol{D}\boldsymbol{A}_{(3)}\boldsymbol{B}\boldsymbol{S}\|_F^2 + \|\boldsymbol{Y}_{(3)} - \boldsymbol{R}\boldsymbol{D}\boldsymbol{A}_{(3)}\|_F^2 + \sum_{k=1}^{K} \lambda \|\mathcal{A}^k\|_{\text{CTNN}}. \tag{18}$$

Apart from the low-rank property in the third mode illustrated in Fig. 3(b), [33] has demonstrated that tensors with different permutation forms also exhibit low-rank characteristics, as illustrated in Fig. 4. Consequently, we apply cascaded transformations across all three tensor modes; rather than restricting them to the third mode alone, the proposed approach reduces sensitivity to the direction that defines lower tubal grades.

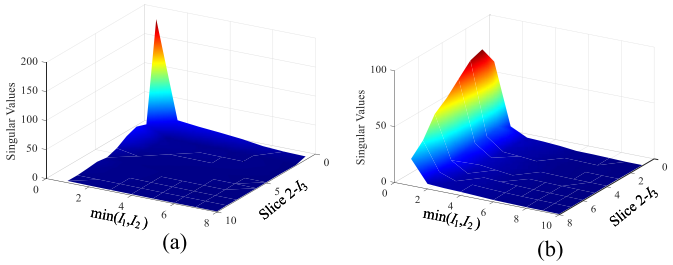


Fig. 4. Plot of singular values based on (a) permute-2 and (b) permute-3.

Definition 15 (Full-Scale CTNN): Given any tensor $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ and cascade transform matrices $\widetilde{\boldsymbol{C}}_k \in \mathbb{R}^{I_k \times I_k}, k = 1, 2, 3$, the FCTNN is achieved by applying cascade transform across all three tensor modes[2]

$$\begin{aligned}
\widetilde{\mathcal{A}} &= \mathcal{A} \times_1 \widetilde{\boldsymbol{C}}_1, \quad \|\mathcal{A}\|_{\text{CTNN}_1} = \sum_{i=1}^{I_1} \|\widetilde{\mathcal{A}}(i,:,:)\|_* \\
\vec{\mathcal{A}} &= \mathcal{A} \times_2 \widetilde{\boldsymbol{C}}_2, \quad \|\mathcal{A}\|_{\text{CTNN}_2} = \sum_{i=1}^{I_2} \|\vec{\mathcal{A}}(:,i,:)\|_* \\
\hat{\mathcal{A}} &= \mathcal{A} \times_3 \widetilde{\boldsymbol{C}}_3, \quad \|\mathcal{A}\|_{\text{CTNN}_3} = \sum_{i=1}^{I_3} \|\hat{\mathcal{A}}(:,:,i)\|_* \\
\|\mathcal{A}\|_{\text{FCTNN}} &= \lambda_1 \|\mathcal{A}\|_{\text{CTNN}_1} + \lambda_2 \|\mathcal{A}\|_{\text{CTNN}_2} \\
&\quad + \lambda_3 \|\mathcal{A}\|_{\text{CTNN}_3}
\end{aligned} \tag{19}$$

where $\widetilde{\mathcal{A}}$, $\vec{\mathcal{A}}$, and $\hat{\mathcal{A}}$ denote the transformed tensors along first, second, and third modes, respectively. $\lambda_{1,2,3}$ are the tradeoff parameters.

Furthermore, to mitigate the excessive penalty on large singular values, this study proposes a more comprehensive log-sum-based FCTNN, which encapsulates essential features, such as edges and corners, and requires less stringent penalization.

Definition 16 (Log-Sum-Based FCTNN): For any $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, the log-sum-based FCTNN (LFCTNN) is defined as follows:

$$\begin{aligned}
\|\mathcal{A}\|_{\text{LFCTNN}} &= \lambda_1 \|\mathcal{A}\|_{\text{LFCTNN}_1} + \lambda_2 \|\mathcal{A}\|_{\text{LFCTNN}_2} \\
&\quad + \lambda_3 \|\mathcal{A}\|_{\text{LFCTNN}_3} \\
&= \frac{1}{I_1} \sum_{i=1}^{I_1} \text{LS}\left(\widetilde{\mathcal{A}}(i,:,:)\right) + \frac{1}{I_2} \sum_{i=1}^{I_2} \text{LS}\left(\vec{\mathcal{A}}(:,i,:)\right) \\
&\quad + \frac{1}{I_3} \sum_{i=1}^{I_3} \text{LS}\left(\hat{\mathcal{A}}(:,:,i)\right) \\
&= \frac{1}{I_1} \sum_{i=1}^{I_1} \sum_j \text{Log}\left(\sigma_j\left(\widetilde{\mathcal{A}}(i,:,:)\right) + \epsilon\right)
\end{aligned}$$

[2]Based on the tensor permute-k operation in Table I, $\|\mathcal{A}\|_{\text{CTNN}_1}$ and $\|\mathcal{A}\|_{\text{CTNN}_2}$ can be converted into their conventional forms along the third dimension, expressed as ipermute$(\sum_{i=1}^{I_1} \|\widetilde{\mathcal{A}}^1(:,i,:)\|_*, 1)$ and ipermute$(\sum_{i=1}^{I_2} \|\vec{\mathcal{A}}^2(:,i,:)\|_*, 2)$, where ipermute denotes the inverse permutation operation.

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$$+\frac{1}{I_2}\sum_{i=1}^{I_2}\sum_{j}\mathrm{Log}\left(\sigma_j\left(\vec{\mathcal{A}}(:,i,:)\right)+\epsilon\right)$$
$$+\frac{1}{I_3}\sum_{i=1}^{I_3}\sum_{j}\mathrm{Log}\left(\sigma_j\left(\hat{\mathcal{A}}(:,:,i)\right)+\epsilon\right) \tag{20}$$

where $\mathrm{LS}(\boldsymbol{A}) = \mathrm{Log}(\sigma_j(\boldsymbol{A})+\epsilon)$, $\sigma_j(\boldsymbol{A})$ denotes the jth singular value of $\boldsymbol{A}$, and ε is a very small positive number.

On that basis, the reduced image estimation problem incorporating LFCTNN and nonlocal spatial constraints can be formulated as follows:

$$\begin{aligned}\min_{\boldsymbol{A}_{(3)}} &\|\boldsymbol{X}_{(3)}-\boldsymbol{D}\boldsymbol{A}_{(3)}\boldsymbol{B}\boldsymbol{S}\|_F^2+\|\boldsymbol{Y}_{(3)}-\boldsymbol{R}\boldsymbol{D}\boldsymbol{A}_{(3)}\|_F^2\\&+\sum_{k=1}^{K}\lambda_1\|\mathcal{A}^k\|_{\mathrm{LFCTNN}_1}+\lambda_2\|\mathcal{A}^k\|_{\mathrm{LFCTNN}_2}\\&+\lambda_3\|\mathcal{A}^k\|_{\mathrm{LFCTNN}_3}.\end{aligned} \tag{21}$$

C. ADMM-Based Solver

To deal with the issue that all optimization variables are coupled with each other, a ADMM-based solver is presented for alternately and recursively optimizing all variables of formula (21), leading to the following unconstrained Lagrangian format:

$$\begin{aligned}&L\left(\mathcal{A},\mathcal{U},\mathcal{V},\mathcal{W},\widetilde{\mathcal{M}},\vec{\mathcal{M}},\hat{\mathcal{M}}\right)\\&=\|\boldsymbol{X}_{(3)}-\boldsymbol{D}\boldsymbol{A}_{(3)}\boldsymbol{B}\boldsymbol{S}\|_F^2+\|\boldsymbol{Y}_{(3)}-\boldsymbol{R}\boldsymbol{D}\boldsymbol{C}_{(3)}\|_F^2\\&+\mu\|\mathcal{U}-\mathcal{A}+\frac{\widetilde{\mathcal{M}}}{2\mu}\|_F^2+\lambda_1\sum_{k=1}^{K}\|\mathcal{U}^k\|_{\mathrm{LFCTNN}_1}\\&+\mu\|\mathcal{V}-\mathcal{A}+\frac{\vec{\mathcal{M}}}{2\mu}\|_F^2+\lambda_2\sum_{k=1}^{K}\|\mathcal{V}^k\|_{\mathrm{LFCTNN}_2}\\&+\mu\|\mathcal{W}-\mathcal{A}+\frac{\hat{\mathcal{M}}}{2\mu}\|_F^2+\lambda_3\sum_{k=1}^{K}\|\mathcal{W}^k\|_{\mathrm{LFCTNN}_3}\end{aligned} \tag{22}$$

where $\mathcal{U}=\mathcal{V}=\mathcal{W}=\mathcal{A}$ are auxiliary variables, $\widetilde{\mathcal{M}}$, $\vec{\mathcal{M}}$, and $\hat{\mathcal{M}}$ denote the Lagrange multipliers, and μ is the penalty parameter. Problem (22) can be addressed by iteratively updating each variable while keeping the others constant. The detail the process for solving each subproblem corresponding to the individual variables involved.

Step 1 (For the $\boldsymbol{A}_{(3)}$-Subproblem): To address the problem, we isolate all terms containing $\boldsymbol{A}$ from (22). The solution requires us to solve the following expression:

$$\begin{aligned}\boldsymbol{A}_{(3)}&\in\underset{\boldsymbol{A}_{(3)}}{\mathrm{argmin}}L\left(\mathcal{A},\mathcal{U},\mathcal{V},\mathcal{W},\widetilde{\mathcal{M}},\vec{\mathcal{M}},\hat{\mathcal{M}}\right)\\&=\underset{\boldsymbol{A}_{(3)}}{\mathrm{argmin}}\|\boldsymbol{X}_{(3)}-\boldsymbol{D}\boldsymbol{A}_{(3)}\boldsymbol{B}\boldsymbol{S}\|_F^2+\|\boldsymbol{Y}_{(3)}-\boldsymbol{R}\boldsymbol{D}\boldsymbol{A}_{(3)}\|_F^2\\&+\mu\|\boldsymbol{U}_{(3)}-\boldsymbol{A}_{(3)}+\frac{\widetilde{\boldsymbol{M}}_{(3)}}{2\mu}\|_F^2+\mu\|\boldsymbol{V}_{(3)}-\boldsymbol{A}_{(3)}+\frac{\vec{\boldsymbol{M}}_{(3)}}{2\mu}\|_F^2\\&+\mu\|\boldsymbol{W}_{(3)}-\boldsymbol{A}_{(3)}+\frac{\hat{\boldsymbol{M}}_{(3)}}{2\mu}\|_F^2.\end{aligned} \tag{23}$$

Equation (23) is a strong convex optimization, which has a sole solution. The derivative of (23) for $\boldsymbol{A}_{(3)}$ is zero when (23) obtains the minimum value, and the solution for (23) satisfies

$$\boldsymbol{H}_1\boldsymbol{A}_{(3)}+\boldsymbol{A}_{(3)}\boldsymbol{H}_2=\boldsymbol{H}_3 \tag{24}$$

where $\boldsymbol{H}_1$ is defined as $[(\boldsymbol{R}\boldsymbol{D})^{\mathrm{T}}\boldsymbol{R}\boldsymbol{D}+\mu\boldsymbol{I}_L]$, which is positive definite with all eigenvalues being positive, with $\boldsymbol{I}_L\in\mathbb{R}^{L\times L}$ being the identity matrix. The matrix $\boldsymbol{H}_2$ is defined as $(\boldsymbol{B}\boldsymbol{S})(\boldsymbol{B}\boldsymbol{S})^{\mathrm{T}}$, which is semipositive. The matrix $\boldsymbol{H}_3$ is given by $(\boldsymbol{R}\boldsymbol{D})^{\mathrm{T}}\boldsymbol{Y}_{(3)}+\boldsymbol{D}^{\mathrm{T}}\boldsymbol{X}_{(3)}(\boldsymbol{B}\boldsymbol{S})^{\mathrm{T}}+\mu(\boldsymbol{U}_{(3)}+(\widetilde{\boldsymbol{M}}_{(3)}/2\mu)+\boldsymbol{V}_{(3)}+(\vec{\boldsymbol{M}}_{(3)}/2\mu)+\boldsymbol{W}_{(3)}+(\hat{\boldsymbol{M}}_{(3)}/2\mu))$. Using the properties of $\boldsymbol{B}$ and $\boldsymbol{S}$, (24) can be solved in a closed form, as detailed in [22] and [45].

Step 2 (For $\mathcal{U},\mathcal{V},\mathcal{W}$-Subproblems): By extracting all terms containing $\mathcal{U}$ from problem (22), we can deduce

$$\begin{aligned}\mathcal{U}&\in\underset{\mathcal{U}}{\mathrm{argmin}}L\left(\mathcal{A},\mathcal{U},\mathcal{V},\mathcal{W},\widetilde{\mathcal{M}},\vec{\mathcal{M}},\hat{\mathcal{M}}\right)\\&=\mu\|\mathcal{U}-\mathcal{A}+\frac{\widetilde{\mathcal{M}}}{2\mu}\|_F^2+\lambda_1\sum_{k=1}^{K}\|\mathcal{U}^k\|_{\mathrm{LFCTNN}_1}.\end{aligned} \tag{25}$$

Problem (25) involves a low-rank tensor regularized optimization approach. We can tackle this issue by processing each cluster individually, i.e.,

$$\underset{\mathcal{U}^k}{\mathrm{argmin}}\lambda_1\sum_{k=1}^{K}\|\mathcal{U}^k\|_{\mathrm{LFCTNN}_1}+\sum_{k=1}^{K}\mu\|\mathcal{U}^k-\mathcal{A}^k+\frac{\widetilde{\mathcal{M}}^k}{2\mu}\|_F^2 \tag{26}$$

where $\mathcal{U}^k$ and $\widetilde{\mathcal{M}}^k$ are tensors formed by the kth cluster patches in $\mathcal{U}$ and $\widetilde{\mathcal{M}}$. Based on the definition of LFCTNN in (20), the kth cluster $\mathcal{U}^k$ can be estimated by solving the following problem:

$$\underset{\widetilde{\mathcal{U}}^k}{\mathrm{minimize}}\quad\frac{\mu}{N_k}\|\widetilde{\mathcal{U}}^k-\mathcal{H}_1^k\|_F^2+\lambda_1\sum_{i=1}^{N_k}\mathrm{LS}\left(\widetilde{\mathcal{U}}^k(i,:,:)\right) \tag{27}$$

where $\widetilde{\mathcal{U}}^k=\mathcal{U}^k\times_1\widetilde{\boldsymbol{C}}_1$ and $\mathcal{H}_1^k=(\mathcal{A}^k-(\widetilde{\mathcal{M}}^k/2\mu))\times_1\widetilde{\boldsymbol{C}}_1$ are transformed tensors. To solve (27), we introduce Theorem 2 named log-sum tensor singular value thresholding (LTSVT). After computing $\widetilde{\mathcal{U}}^k$, the inverse cascade transform with the first mode is applied: $\mathcal{U}^k=\widetilde{\mathcal{U}}^k\times_1\widetilde{\boldsymbol{C}}_1^{-1}$. Finally, $\mathcal{U}$ is obtained by aggregating the clusters $\widetilde{\mathcal{U}}^k$ for $1\le K$.

Theorem 2 (Log-Sum Tensor Singular Value Thresholding): Given any tensor $\mathcal{X}\in\mathbb{R}^{I_1\times I_2\times I_3}$, a minimization problem for LTSVT regularization can be formulated as

$$\arg\min_{\mathcal{X}}\frac{1}{2}\|\mathcal{X}-\mathcal{Y}\|_F^2+\tau\sum_{i=1}^{I_1}\mathrm{LS}(\mathcal{X}(i,:,:)) \tag{28a}$$
$$\arg\min_{\mathcal{X}}\frac{1}{2}\|\mathcal{X}-\mathcal{Y}\|_F^2+\tau\sum_{i=1}^{I_2}\mathrm{LS}(\mathcal{X}(:,i,:)) \tag{28b}$$
$$\arg\min_{\mathcal{X}}\frac{1}{2}\|\mathcal{X}-\mathcal{Y}\|_F^2+\tau\sum_{i=1}^{I_3}\mathrm{LS}(\mathcal{X}(:,:,i)). \tag{28c}$$

The optimal solution of problem (28) can be indicated by the following LTSVT operator:

$$\mathrm{LTSVT}_{\tau,\varepsilon}(\mathcal{Y})=\mathcal{U}*\mathcal{D}_{\tau,\varepsilon}*\mathcal{V}^{\mathrm{T}}. \tag{29}$$

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$\mathcal{D}_{\tau,\varepsilon}(\cdot)$ is the thresholding operator given by

$$\mathcal{D}_{\tau,\varepsilon}(i,:,:)=\begin{cases}0, & \text{if } c_2\le 0\\ \text{sign}(\mathcal{S}(i,:,:))\left(\dfrac{c_1+\sqrt{c_2}}{2}\right), & \text{if } c_2>0\end{cases} \tag{30a}$$

$$\mathcal{D}_{\tau,\varepsilon}(:,i,:)=\begin{cases}0\\ \text{sign}(\mathcal{S}(:,i,:))\left(\dfrac{c_1+\sqrt{c_2}}{2}\right)\end{cases} \tag{30b}$$

$$\mathcal{D}_{\tau,\varepsilon}(:,:,i)=\begin{cases}0\\ \text{sign}(\mathcal{S}(:,:,i))\left(\dfrac{c_1+\sqrt{c_2}}{2}\right).\end{cases} \tag{30c}$$

As an example, consider (30a), where $\mathcal{S}(i,:,:)$ is the ith horizontal slice, $c_1=|\mathcal{S}(i,:,:)|-\varepsilon$, and $c_2=(\mathcal{S}(i,:,:)+\varepsilon)^2-4\tau$. Similarly, c_1 and c_2 in (30b) and (30c) correspond to the respective permuted states of $\mathcal{S}$.

The solution to the subproblems for $\mathcal{V}$ and $\mathcal{W}$ is analogous to that for $\mathcal{U}$, with the distinction arising from the variations $(i,:,:)$ in Theorem 2. Due to rotational properties, the solutions for $\mathcal{V}$ and $\mathcal{W}$ are with $\mathcal{V}(:,i,:)$ and $\mathcal{W}(:,:,i)$, respectively.

Step 3 (Update of the Lagrangian Parameters): Under the principle of general ADMM, we can update the Lagrangian multipliers in parallel as

$$\begin{aligned}\widetilde{\mathcal{M}}&=\widetilde{\mathcal{M}}+2\mu(\mathcal{U}-\mathcal{A})\\ \vec{\mathcal{M}}&=\vec{\mathcal{M}}+2\mu(\mathcal{V}-\mathcal{A})\\ \hat{\mathcal{M}}&=\hat{\mathcal{M}}+2\mu(\mathcal{W}-\mathcal{A}).\end{aligned} \tag{31}$$

Summarizing the aforementioned descriptions, we present the entire ADMM-based solution process for solving problem (22) in Algorithm 1.

Algorithm 1 ADMM for Solving (22)

Input: LR-HSI $\mathcal{Y}$; HR-MSI $\mathcal{Z}$; the parameter μ; Cluster number K; path size p; step size s, Max iteration number T.

Output: The target HR-HSI $\mathcal{X}$.

Initialization:$\mu=1e-3$, $K=400$, $p=6(8)$, $s=4$, $T=30$;
while $t\leqslant T$ **do**
  Update $\boldsymbol{A}_{(3)}$ via (24);
  Update $\mathcal{U},\mathcal{V},\mathcal{W}$ via **Theorem** 2;
  Update $\widetilde{\mathcal{M}},\vec{\mathcal{M}},\hat{\mathcal{M}}$ via (31);
  $t\leftarrow t+1$;
end

D. Computational Complexity

For the acquired LRHSI $\mathcal{Y}\in\mathbb{R}^{w\times h\times S}$ and the corresponding HRMSI $\mathcal{Z}\in\mathbb{R}^{W\times H\times s}$, the per-iteration cost is primarily determined by updating $\boldsymbol{A}_{(3)}\in\mathbb{R}^{L\times HW}$ and solving the subproblems for $\mathcal{U},\mathcal{V},\mathcal{W}$ based on the group $\mathcal{A}^k\in\mathbb{R}^{b\times L\times N_k}$. In each iteration, the computational complexity of updating $\boldsymbol{A}_{(3)}$ is $\mathcal{O}(HW\max\{\log(HW),L^2\})$. In addition, solving the subproblems for $\mathcal{U},\mathcal{V},\mathcal{W}$ involves (inverse) double-domain transformations and SVDs of all frontal slices. Therefore, the time complexities for solving $\mathcal{U}$, $\mathcal{V}$, and $\mathcal{W}$ are $\mathcal{O}(\text{bLN}_k(N_k+\min(b,L))+b^2)$, $\mathcal{O}(\text{bLN}_k(b+\min(N_k,L))+L^2)$, and $\mathcal{O}(\text{bLN}_k(L+\min(N_k,b))+N_k^2)$, respectively.

IV. Experiments

For the evaluation of LFCTNN-FC and LFCTNN-FL, several cutting-edge approaches are selected as competitors, including coupled sparse tensor factorization (CSTF)[3] [21], low-rank tensor-approximation (LRTA)[4] [46], low tensor multirank regularization (LTMR)[5] [22], coupled tensor ring factorization (CTRF)[6] [47], external and internal prior learning-based method (ExInL)[7] [48], and generalized TNN-based approach (GTNN)[8] [25]. To guarantee a fair comparison, all the unspecified parameters for the methods under comparison are optimally tuned to achieve their best performance. The experiments are conducted using MATLAB R2020a on a Windows 10 system, with an AMD R7-5800 processor and 16 GB of memory.

The practical experiments are conducted on three simulated datasets: PU [49], Washington DC Mall (WDC) [50], and Columbia Computer Vision Laboratory (CAVE) [51]. For the first two datasets, the LRHSI is obtained by applying a 7×7 Gaussian blur with a standard deviation of 2, followed by downsampling by a factor of 4. An IKONOS-like spectral response filter [52] is used to generate the LRHSI. The HSI and MSI are then corrupted with Gaussian noise, achieving signal-to-noise ratios (SNRs) of 30 and 35 dB, respectively. For the CAVE dataset, the spectral response of a Nikon D700 camera [53] is utilized for spectral downsampling, resulting in HRMSI. The LRHSI are generated by applying uniform blur with a ratio of 16, while Gaussian noise is introduced to both the HSI and MSI, each maintaining an SNR of 20 dB. Moreover, to perform realistic HSISR, we utilize a Sentinel-2 image as the HR-MSI and data from the Hyperion sensor aboard the Earth Observing-1 satellite as the LR-HSI [2]. Details of the dataset are provided in the Supplementary Material.

A. Quantitative Metrics

To comprehensively assess the quantitative results, six widely used metrics, i.e., peak SNR (PSNR) [44], structural similarity (SSIM) [54], spectral angle mapper (SAM) [55], Erreur Relative Globale Admensionnelle de Synthese (ERGSA) [56], the degree of distortion (DD) [21], and cross correlation (CC) [57], are adopted. Among these metrics, higher values of PSNR, SSIM, and CC indicate better fusion results, while lower values of SAM, ERGAS, and DD correspond to better fusion outcomes.

[3]https://github.com/renweidian/CSTF
[4]https://openremotesensing.net/knowledgebase/hyperspectral-restoration-and-fusion-with-multispectral-imagery-by-recasting-low-rank-tensor-approximation/
[5]https://github.com/renweidian/LTMR
[6]https://prowdiy.github.io/weihe.github.io/publication.html
[7]https://github.com/renweidian/ExInL
[8]https://github.com/renweidian/GTNN

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B. Synthetic Data Experiments

In this section, we exhibit the generated results of all the competing methods on the three selected datasets, i.e., PU, WDC, and 32 CAVE datasets. The best values are highlighted in red the second-best in blue. For CAVE, the values represent the average across 32 datasets.

1) Quantitative Analysis: Numerical results generated from all the competitors are presented in Table II, the best values and the second best values are respectively highlighted by red and blue colors. For the PU dataset, the proposed LFCTNN-FL generates the best results compared to other competing approaches in terms of all quantitative metrics. Specifically, the PSNR achieved by LFCTNN-FL is 43.4734 dB, marking a significant increase of approximately 3.58% over the CSTF method and 1.25% over GTNN, underscoring the superior capability in preserving high image quality. In addition, the ERGAS value of LFCTNN-FL stands at 1.0233, marking a substantial enhancement of 6.44% relative to GTNN and 12.20% over ExInL, indicating more accurate and efficient image reconstruction. Collectively, these results establish the LFCTNN-FL method as not only outperforming existing techniques but also significantly advancing the state-of-the-art in tensor-based image reconstruction for the PU dataset. The proposed LFCTNN-FC closely follows, with a PSNR of 43.3575 dB, only 0.27% lower than LFCTNN-FL, yet still markedly superior to other competitors. This near parity between LFCTNN-FC and LFCTNN-FL suggests that both proposed methods effectively capture low-rank tensor structures, with LFCTNN-FL exhibiting a slight overall advantage.

For the WDC dataset, the proposed LFCTNN-FL and LFCTNN-FC exhibit notable strengths across various metrics, further establishing their robustness and effectiveness. The improvement is particularly evident in the PSNR metric, where LFCTNN-FL surpasses other approaches such as CSTF, LRTA, LTMR, CTRF, ExInL, and GTNN by margins of 1.86, 4.09, 1.28, 1.97, 0.62, and 0.43 dB, respectively. This improvement in PSNR underscores the superior capability of LFCTNN-FL in leading high-performed image quality. In addition, LFCTNN-FC demonstrates the highest SSIM value of 0.9851, indicating superior SSIM compared to others. The complementary strengths of both methods suggest that they offer substantial advancements over existing state-of-the-art techniques, providing more comprehensive solutions for tensor-based image reconstruction in the WDC dataset.

Upon reviewing the CAVE dataset results, it can be observed that LFTCNN-FL performs exceptionally well on most metrics. Numerically, LFTCNN-FL achieves the highest PSNR of 35.4321 dB, highlighting its excellent ability to preserve image fidelity. In addition, it obtains the lowest ERGAS value of 1.3838 and the lowest DD value of 3.6982, indicating superior spectral and spatial accuracy. LFTCNN-FC closely follows, also exhibiting strong image fusion capabilities. Overall, the superiority of both proposals, LFTCNN-FL, and LFTCNN-FC, underscores that linking transformation domains can enhance low rankness, making it more effective at exploring underlying information.

To further demonstrate the superiority of our proposed method on quantitative effect, Fig. 5 numerically shows bandwise results on representative data including PU and *the fake and real peppers*. It is obvious that the curves of LFCTNN-FL are evidently superior to others in almost all bands, which, together with the previous analysis, illustrates the advantages of our method both spatially and spectrally.

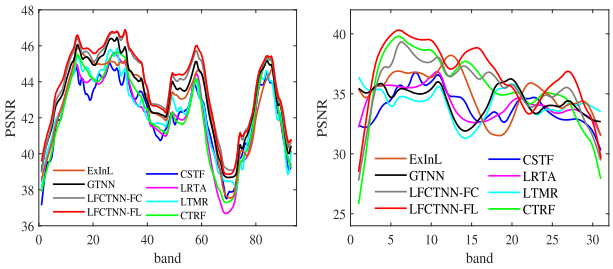


Fig. 5. PSNR values versus different bands on (a) PU and (b) *the fake and real peppers*.

2) Qualitative Analysis: To visually compare the performance of LFTCNN-FC and LFTCNN-FL against other methods, Figs. 6 and 7 present a series of reconstructed images alongside their corresponding error maps of WDC, respectively. For enhanced visual clarity, representative regions of these images have been magnified. It is important to note that the error map illustrates the discrepancies between the fusion results and the ground truth, where darker blue regions indicate fewer residuals. At first glance, the green road in the LRTA reconstruction appears uneven and lacks smoothness. Upon closer inspection, the magnified inset reveals significant green contamination on the roof area in the LRTA output; the first roof displays three distinct green streaks, while the second shows three unrelated green spots. Moving to LTMR, CTRF, ExInL, and GTNN, these methods demonstrate moderate enhancements in both clarity and detail retention. However, their error maps still reveal substantial inaccuracies, particularly with scattered high-intensity errors in areas that should otherwise be homogeneous. The error map for LFCTNN-FL is characterized by fewer and less intense error regions, which suggests that this method successfully balances detail preservation with smoothness across the image. Notably, the proposed LFCTNN-FC tends to yield results that are characterized by a smoother and bluer appearance, which indicates that this method excels in preserving fine details while avoiding common artifacts such as oversmoothing or excessive noise, thereby producing the most accurate and visually appealing results in the comparison.

For the CAVE *superballs* dataset depicted in the upper figure of Fig. 8, the CSTF method produces reconstructed surfaces of the spheres with noticeable artifacts and noise. Upon closer inspection, the enlarged inset reveals blurred texture details and compromised color uniformity, indicating limitations in capturing fine details and maintaining color fidelity. The LRTA method displays evident texture distortions and color contamination, particularly noticeable in the reconstruction of the spheres, where inconsistencies are prominent. The magnified regions reveal visible distortions and color bleed. LTMR and CTRF yield relatively smoother reconstructions with more accurate color rendering than LRTA. However, it still lacks detail sharpness on the surface of the spheres, and noise contamination is apparent. The ExInL, GTNN, and LFCTNN-

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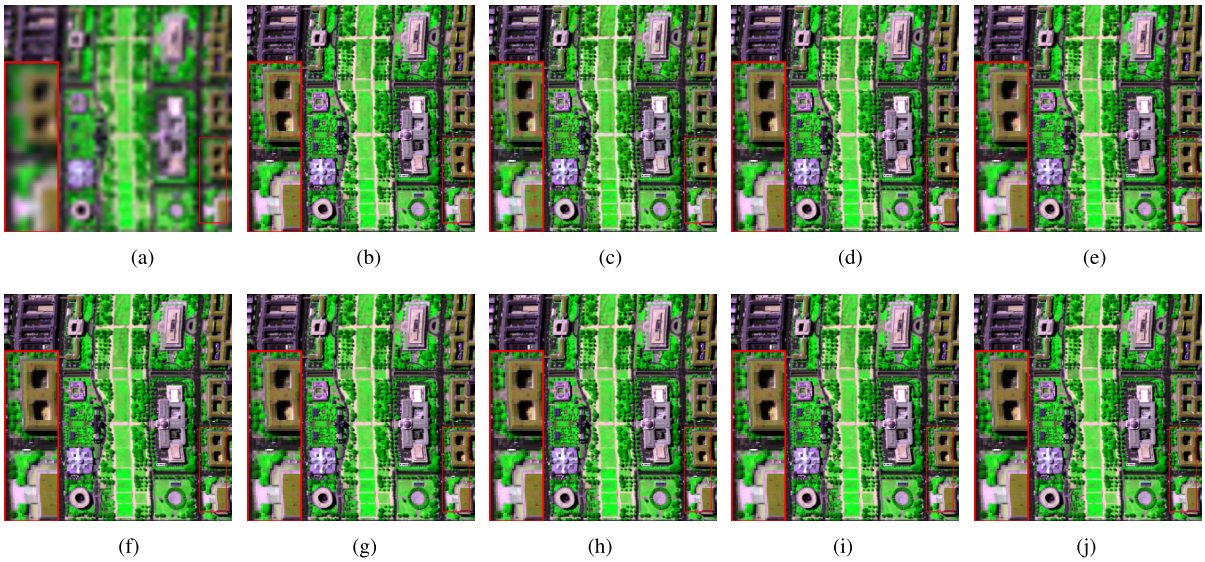


Fig. 6. Pseudocolor images (50th, 70th, and 30th bands) on WDC. (a) LRHSI. (b) CSTF [21]. (c) LRTA [46]. (d) LTMR [22]. (e) CTRF [47]. (f) ExInL [48]. (g) GTNN [25]. (h) LFCTNN-FC. (i) LFCTNN-FL. (j) Ground truth.

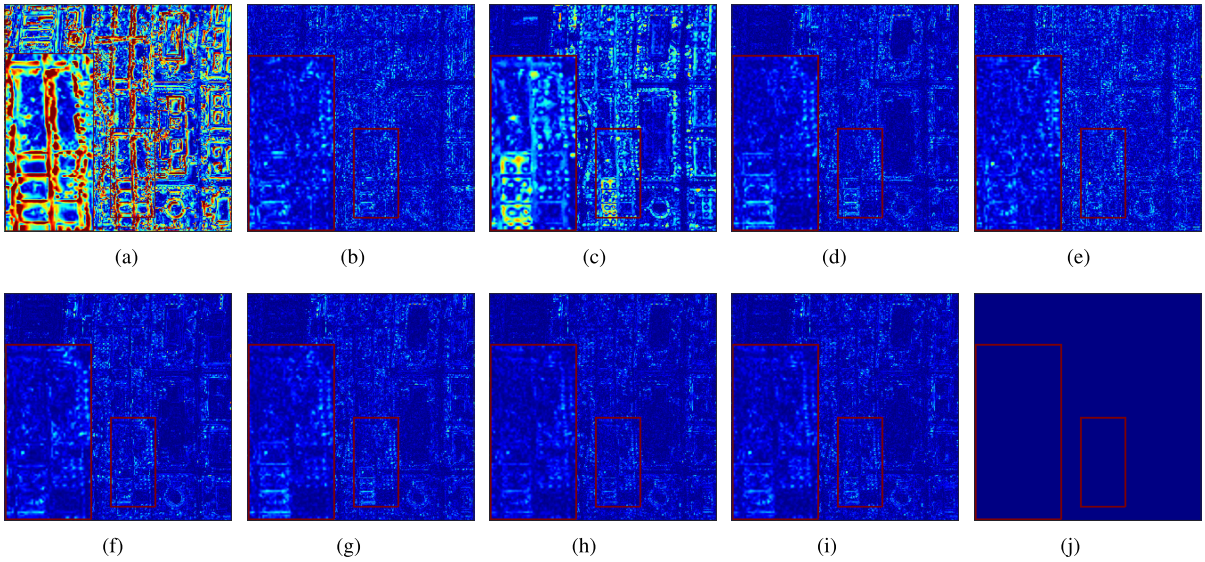


Fig. 7. Error images on WDC. (a) LRHSI. (b) CSTF [21]. (c) LRTA [46]. (d) LTMR [22]. (e) CTRF [47]. (f) ExInL [48]. (g) GTNN [25]. (h) LFCTNN-FC. (i) LFCTNN-FL. (j) Ground truth.

FL methods demonstrate improvements in color reproduction and detail preservation compared to earlier approaches; however, some blurring and loss of fine textures remain, and the glossiness of the surfaces is not accurately rendered. Among the methods, LFCTNN-FC achieves outstanding reconstruction quality, with preserved details and color fidelity nearly identical to the ground truth. The images are clear, with accurate colors and well-preserved textures. The spheres exhibit smooth surfaces with natural glossiness.

For *the fake and real peppers* dataset in the lower figure of Fig. 9, the CSTF and LRTA methods struggle significantly with detail reconstruction, exhibiting pronounced color contamination and loss of detail. CTRF performs moderately, with some blurriness on the surfaces of the fruit and geometric shapes, indicating suboptimal detail preservation. The magnified areas show noticeable noise, especially in high-contrast regions. LTMR and GTNN show improved color reproduction; however, they still lack sharpness in both details and edges. Zoomed-in areas display good color uniformity, but reflective details remain slightly blurred. ExInL exhibits strong denoising capabilities though there remains room for improvement in terms of fine detail preservation. LFCTNN-FC achieves near-perfect reconstruction results, retaining rich details and accurate colors. The performance in high-light and reflective areas is particularly impressive, with no noticeable artifacts. LFCTNN-FL also performs exceptionally well, producing reconstructions nearly identical to LFCTNN-FC, with sharp details and realistic color reproduction.

C. Results on Real Data

Since the above experiments are implemented on simulated HSI and MSI, we use the real HSI and MSI to further test

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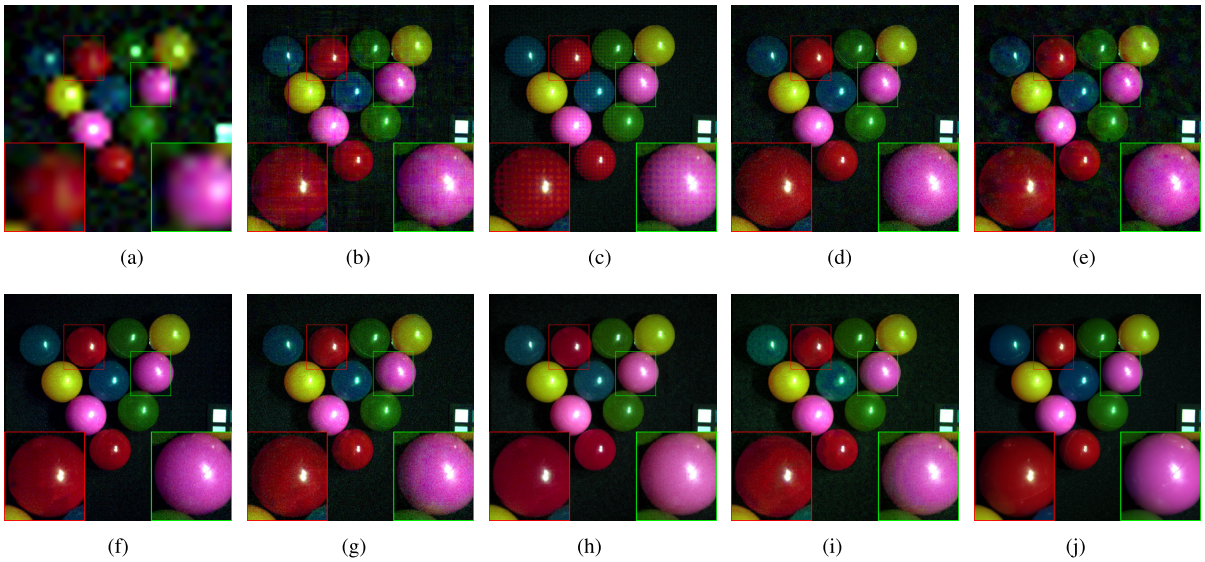


Fig. 8. Pseudocolor images (20th, tenth, and fifth bands) on the *superballs*. (a) LRHSI. (b) CSTF [21]. (c) LRTA [46]. (d) LTMR [22]. (e) CTRF [47]. (f) ExInL [48]. (g) GTNN [25]. (h) LFCTNN-FC. (i) LFCTNN-FL. (j) Ground truth.

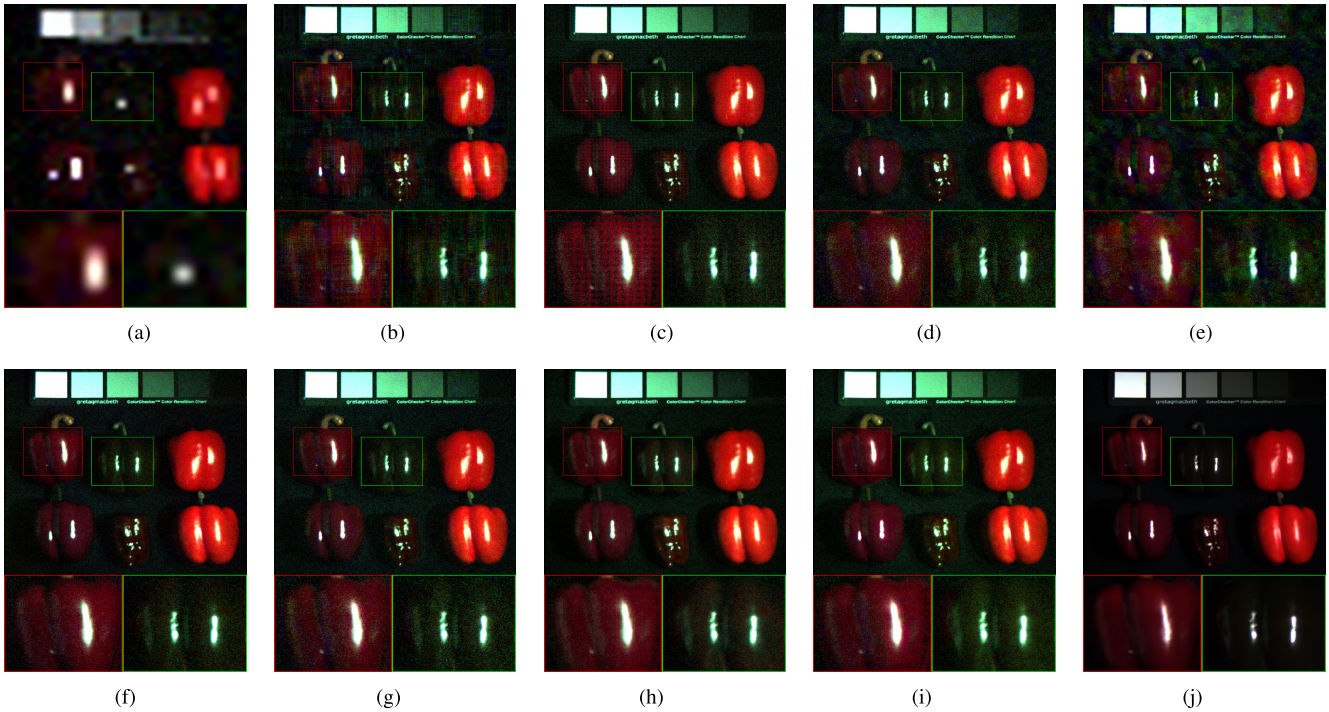


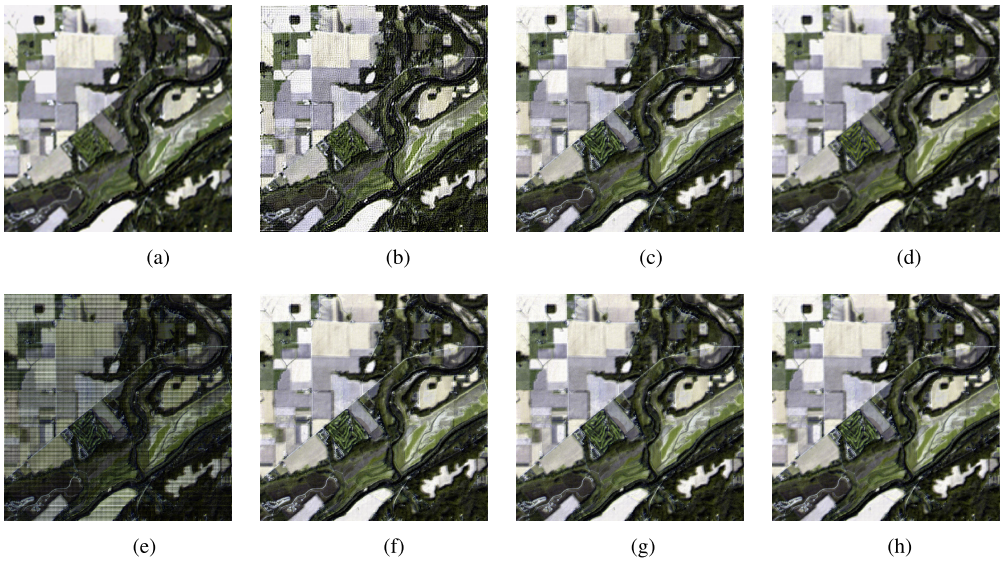
Fig. 9. Pseudocolor images (20th, tenth, and fifth bands) on the *the fake and real peppers*. (a) LRHSI. (b) CSTF [21]. (c) LRTA [46]. (d) LTMR [22]. (e) CTRF [47]. (f) ExInL [48]. (g) GTNN [25]. (h) LFCTNN-FC. (i) LFCTNN-FL. (j) Ground truth.

the compared methods. In the absence of the ground-truth HSI, visual results can demonstrate the superiority of the proposed methods for real HSI super-resolution. Fig. 10 shows the super-resolved results of the compared nonblind methods on the real dataset. Among all competitors, CSTF and ExInL fail to achieve a clear target, whose generated results are full of vagueness. Besides, though CTRF can produce satisfactory spatial details, these methods have spectral distortions in some areas when taking the LR-HSI as the spectral quality reference. LTMR, GTNN, and the proposed LFCTNN-FC/FU perform better in keeping the spatial details of the image.

D. Parameters' Analysis

1) Influence of Low-Rank Tensor Regularization: The LFCTNN approach is grounded in the proposed low-rank tensor regularization, making the parameters λ_1, λ_2, and λ_3 crucial, as they dictate the extent of the low rankness in the recovered tensors. Fig. 11(a) and (b) presents the PSNR curves corresponding to different values of λ_1, λ_2, and λ_3 for LFCTNN-FC and LFCTNN-FL on the WDC dataset, respectively. For simplicity, we assume that λ_1 and λ_3 are equal to reduce the cumbersome process of parameter

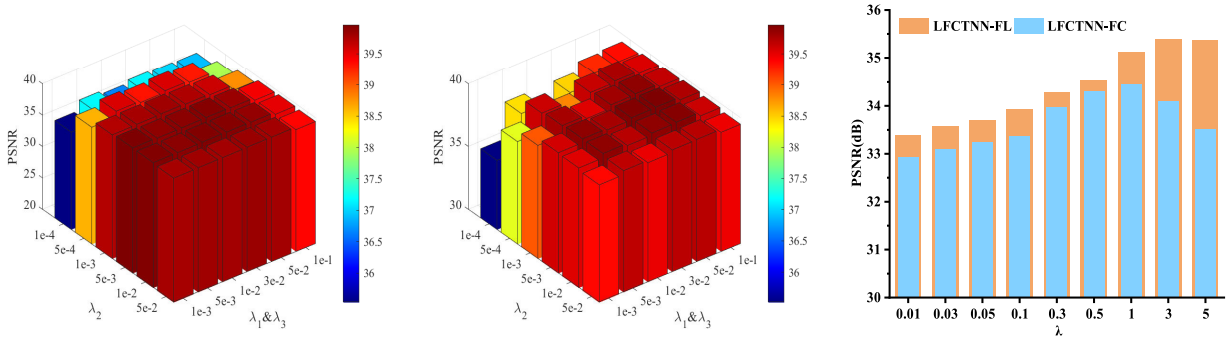
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(a) (b) (c) (d)

(e) (f) (g) (h)

Fig. 10. Pseudocolor images (tenth, ninth, and first bands) in real HSI and MSI fusion. (a) LRHSI. (b) CSTF [21]. (c) LTMR [22]. (d) CTRF [47]. (e) ExInL [48]. (f) GTNN [25]. (g) LFCTNN-FC. (h) LFCTNN-FL.



(a) (b) (c)

Fig. 11. PSNR value versus different selections of λ_1, λ_2, and λ_3 on (a) WDC using LFCTNN-FC, (b) WDC using LFCTNN-FL, and (c) CAVE using both LFCTNN-FC and LFCTNN-FL.

selection. From Fig. 11(a), we observe that PSNR increases as λ_2 grows, reaching its peak at $5e-3$. In contrast, λ_1 and λ_3 demonstrate relatively stable low-rank capability with minimal data fluctuation. Consequently, we suggest a preliminary attempt on pairs [$3e-2$,$5e-3$] for LFCTNN-FC. As for the LFCTNN-FL in Fig. 11(b), one can observe that the experimental results have a sharp rise as parameter λ_2 varies from $1e-4$ to $1e-2$. However, when the peak value is surpassed, the performance declines rapidly as λ_2 grows further. λ_1 and λ_3 show a similar trend to that of LFCTNN-FC, and thus, we adopt the same parameter settings. In summary, the recommended parameter settings for LFCTNN-FL are λ_1, $\lambda_3 = 3e-2$, and $\lambda_2 = 1e-2$.

For the CAVE datasets, we assume that λ_1, λ_2, and λ_3 for different statements are equal, i.e., $\lambda_1 = \lambda_2 = \lambda_3$. Fig. 11(c) illustrates that the orange bars for FU exhibit a steady increase as lambda increases, peaking at $\lambda = 3$ before showing a gradual decline. Similarly, the blue bars for FC reach their maximum PSNR at $\lambda = 1$. Therefore, we recommend setting the λ values for LFCTNN-FL and LFCTNN-FC to 3 and 1, respectively.

2) Analysis of Atom Selection for Parameter L: The parameter L controls the reduced spatial dimension. In Fig. 12, we present PSNR curves as a function of L for the PU, WDC, and CAVE datasets, respectively. In Fig. 12(a), we observe an upward trend in PSNR values as L increases from 4 to 8. However, beyond this peak, performance begins to decline as L further increases. The WDC dataset in Fig. 12(b) follows a similar pattern, with the optimal turning point for L in LFCTNN-FC occurring at 10, while the peak for LFCTNN-FL appears at 12. To streamline parameter tuning, we recommend setting L uniformly to 10.

In Fig. 12(c), the CAVE dataset demonstrates that $L = 4$ is optimal for LFCTNN-FC, whereas the best performance for LFCTNN-FL is achieved at $L = 6$. For consistency across parameters, we set L to 4 in this case as well.

E. Ablations and Discussions

1) Convergence Behavior: The numerical convergence of the ADMM-based solver for our model is evaluated empirically. We compute the relative error ($\|\mathcal{X}^{t+1} - \mathcal{X}^t\|_F / \|\mathcal{X}^t\|_F$)

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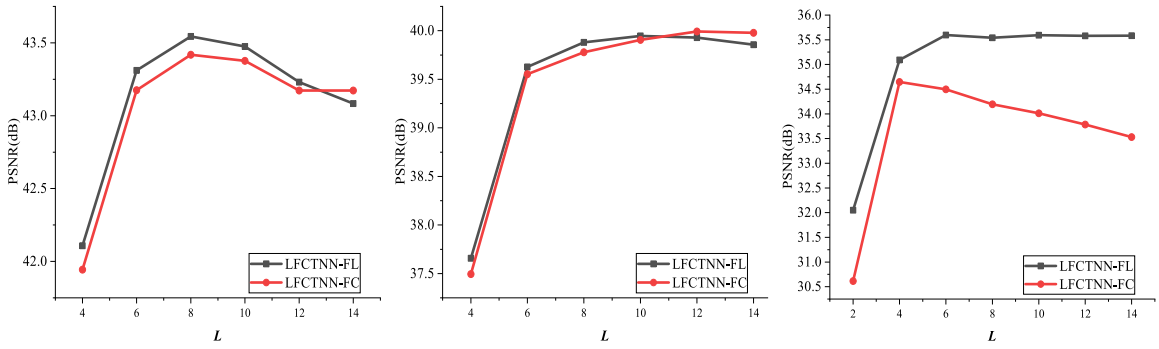


Fig. 12. PSNR values versus different selections of subspace dimension L on (a) PU, (b) WDC, and (c) *cd*.

TABLE II

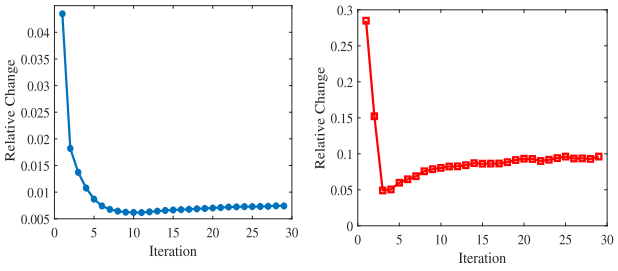
QUANTITATIVE METRICS (PSNR, SSIM, SAM, ERGAS, DD, AND CC) OF ALL COMPARED METHODS. THE BEST VALUES AND THE SECOND BEST VALUES ARE, RESPECTIVELY, HIGHLIGHTED IN RED AND BLUE

Image	Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	DD↓	CC↑
Pavia University	$\text{CSTF}_{\text{TCYB22}}$	$41.9193_{\downarrow 3.57\%}$	$0.9809_{\downarrow 0.61\%}$	$2.4010_{\uparrow 24.86\%}$	$1.2369_{\uparrow 20.87\%}$	$1.5241_{\uparrow 24.52\%}$	$0.9962_{\downarrow 0.11\%}$
	$\text{LRTA}_{\text{TGRS21}}$	$41.9203_{\downarrow 3.57\%}$	$0.9820_{\downarrow 0.50\%}$	$2.3098_{\uparrow 20.12\%}$	$1.2300_{\uparrow 20.20\%}$	$1.5064_{\uparrow 23.07\%}$	$0.9961_{\downarrow 0.12\%}$
	$\text{LTMR}_{\text{TIP22}}$	$42.1415_{\downarrow 3.06\%}$	$0.9803_{\downarrow 0.67\%}$	$2.3780_{\uparrow 23.67\%}$	$1.9270_{\uparrow 88.31\%}$	$1.5003_{\uparrow 22.57\%}$	$0.9972_{\downarrow 0.01\%}$
	$\text{CTRF}_{\text{RP22}}$	$41.9957_{\downarrow 3.40\%}$	$0.9813_{\downarrow 0.57\%}$	$2.3309_{\uparrow 21.22\%}$	$1.2344_{\uparrow 20.63\%}$	$1.4925_{\uparrow 21.94\%}$	$0.9961_{\downarrow 0.12\%}$
	$\text{ExInL}_{\text{SCIS23}}$	$42.5156_{\downarrow 2.20\%}$	$0.9861_{\downarrow 0.08\%}$	$2.0039_{\uparrow 4.21\%}$	$1.1481_{\uparrow 12.20\%}$	$1.3227_{\uparrow 8.06\%}$	$0.9967_{\downarrow 0.06\%}$
	$\text{GTNN}_{\text{TNNLS24}}$	$42.9282_{\downarrow 1.25\%}$	$0.9846_{\downarrow 0.23\%}$	$2.0943_{\uparrow 8.91\%}$	$1.0892_{\uparrow 6.44\%}$	$1.3309_{\uparrow 8.73\%}$	$0.9970_{\downarrow 0.03\%}$
	LFCTNN-FC	$43.3575_{\downarrow 0.27\%}$	$0.9866_{\downarrow 0.03\%}$	$1.9494_{\uparrow 1.38\%}$	$1.0419_{\uparrow 1.82\%}$	$1.2413_{\uparrow 1.41\%}$	$0.9973_{\downarrow 0.00\%}$
	LFCTNN-FL	$43.4734_{\downarrow 0.00\%}$	$0.9869_{\downarrow 0.00\%}$	$1.9229_{\uparrow 0.00\%}$	$1.0233_{\uparrow 0.00\%}$	$1.2240_{\uparrow 0.00\%}$	$0.9973_{\downarrow 0.00\%}$
Washington DC Mall	$\text{CSTF}_{\text{TCYB22}}$	$38.0893_{\downarrow 4.65\%}$	$0.9769_{\downarrow 0.84\%}$	$4.4355_{\uparrow 34.30\%}$	$1.6415_{\uparrow 24.38\%}$	$2.7366_{\uparrow 32.23\%}$	$0.9948_{\downarrow 0.18\%}$
	$\text{LRTA}_{\text{TGRS21}}$	$35.8487_{\downarrow 10.26\%}$	$0.9642_{\downarrow 2.13\%}$	$5.2077_{\uparrow 57.68\%}$	$2.3112_{\uparrow 75.13\%}$	$3.5685_{\uparrow 72.43\%}$	$0.9903_{\downarrow 0.63\%}$
	$\text{LTMR}_{\text{TIP22}}$	$38.6632_{\downarrow 3.21\%}$	$0.9805_{\downarrow 0.48\%}$	$3.9095_{\uparrow 18.37\%}$	$1.5161_{\uparrow 14.88\%}$	$2.4607_{\uparrow 18.90\%}$	$0.9956_{\downarrow 0.10\%}$
	$\text{CTRF}_{\text{RP22}}$	$37.9790_{\downarrow 4.92\%}$	$0.9765_{\downarrow 0.86\%}$	$4.5154_{\uparrow 36.71\%}$	$1.6741_{\uparrow 26.85\%}$	$2.7968_{\uparrow 35.14\%}$	$0.9946_{\downarrow 0.20\%}$
	$\text{ExInL}_{\text{SCIS23}}$	$39.3254_{\downarrow 1.55\%}$	$0.9846_{\downarrow 0.06\%}$	$3.3029_{\uparrow 0.00\%}$	$1.3923_{\uparrow 5.50\%}$	$2.1871_{\uparrow 5.68\%}$	$0.9964_{\downarrow 0.02\%}$
	$\text{GTNN}_{\text{TNNLS24}}$	$39.5127_{\downarrow 1.08\%}$	$0.9842_{\downarrow 0.10\%}$	$3.4250_{\uparrow 3.70\%}$	$1.3585_{\uparrow 2.94\%}$	$2.1967_{\uparrow 6.15\%}$	$0.9965_{\downarrow 0.01\%}$
	LFCTNN-FC	$39.9206_{\downarrow 0.06\%}$	$0.9851_{\downarrow 0.00\%}$	$3.3955_{\uparrow 2.81\%}$	$1.3295_{\uparrow 0.74\%}$	$2.0844_{\uparrow 0.72\%}$	$0.9966_{\downarrow 0.00\%}$
	LFCTNN-FL	$39.9451_{\downarrow 0.00\%}$	$0.9850_{\downarrow 0.02\%}$	$3.3028_{\uparrow 0.00\%}$	$1.3197_{\uparrow 0.00\%}$	$2.0695_{\uparrow 0.00\%}$	$0.9966_{\downarrow 0.00\%}$
CAVE	$\text{CSTF}_{\text{TCYB22}}$	$33.2476_{\downarrow 6.17\%}$	$0.7563_{\downarrow 13.12\%}$	$22.1949_{\uparrow 16.73\%}$	$1.5932_{\uparrow 18.91\%}$	$4.5818_{\uparrow 23.89\%}$	$0.9798_{\downarrow 0.71\%}$
	$\text{LRTA}_{\text{TGRS21}}$	$33.8913_{\downarrow 4.35\%}$	$0.7572_{\downarrow 13.02\%}$	$21.5351_{\uparrow 13.26\%}$	$1.4671_{\uparrow 6.02\%}$	$4.3794_{\uparrow 18.42\%}$	$0.9827_{\downarrow 0.42\%}$
	$\text{LTMR}_{\text{TIP22}}$	$33.7787_{\downarrow 4.67\%}$	$0.7557_{\downarrow 13.19\%}$	$20.4871_{\uparrow 7.75\%}$	$1.5324_{\uparrow 10.74\%}$	$4.3532_{\uparrow 17.71\%}$	$0.9841_{\downarrow 0.27\%}$
	$\text{CTRF}_{\text{PR22}}$	$34.6299_{\downarrow 2.26\%}$	$0.8259_{\downarrow 5.12\%}$	$20.8123_{\uparrow 9.46\%}$	$1.4830_{\uparrow 7.23\%}$	$4.1655_{\uparrow 12.64\%}$	$0.9775_{\downarrow 0.94\%}$
	$\text{ExInL}_{\text{SCIS23}}$	$34.2481_{\downarrow 3.34\%}$	$0.7756_{\downarrow 10.90\%}$	$19.0136_{\uparrow 0.00\%}$	$1.4399_{\uparrow 4.05\%}$	$4.2138_{\uparrow 13.94\%}$	$0.9829_{\downarrow 0.40\%}$
	$\text{GTNN}_{\text{TNNLS24}}$	$33.9794_{\downarrow 4.10\%}$	$0.7691_{\downarrow 11.65\%}$	$20.0672_{\uparrow 5.54\%}$	$1.4727_{\uparrow 6.42\%}$	$4.2454_{\uparrow 14.80\%}$	$0.9843_{\downarrow 0.25\%}$
	LFCTNN-FC	$35.0560_{\downarrow 1.06\%}$	$0.8394_{\downarrow 3.57\%}$	$25.5789_{\uparrow 34.53\%}$	$1.4120_{\uparrow 2.04\%}$	$3.8515_{\uparrow 4.15\%}$	$0.9848_{\downarrow 0.20\%}$
	LFCTNN-FL	$35.4321_{\downarrow 0.00\%}$	$0.8705_{\downarrow 0.00\%}$	$24.6445_{\uparrow 29.62\%}$	$1.3838_{\uparrow 0.00\%}$	$3.6982_{\uparrow 0.00\%}$	$0.9868_{\downarrow 0.00\%}$

between successively recovered images, with the results shown in Fig. 13. The curves indicate that the relative error decreases sharply following an initial fluctuation. As the number of iterations increases, the relative error approaches a stable value, suggesting that numerical convergence is achieved.

2) Results on Running Time: Running efficiency is also an important metric for the practical applications of newly designed models. Table III reports the computational time of all test methods on PU, WDC, *balloons*, and *cd*, respectively. Evidently, due to their simple model architectures and update rules, ExInL, LRTA, and LTMR demonstrate the highest computational efficiency among all competing methods. In contrast, the LFCTNN approach exhibits relatively low computational efficiency, primarily due to the cost associated with processing the three modes of the input tensor to explore low-tensor rank properties. In addition, LFCTNN-FL incurs further computational overhead as it performs an SVD operation in each iteration to obtain the adaptive transformation

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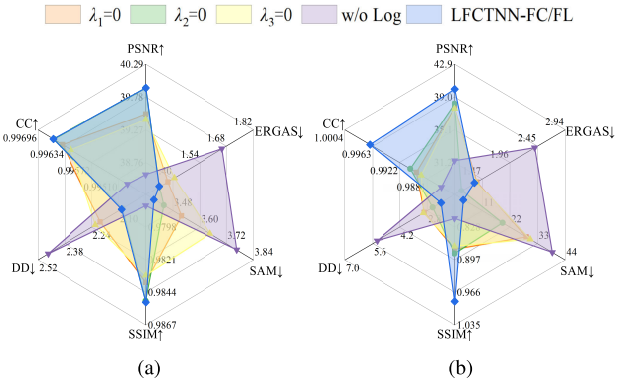


(a) (b)

Fig. 13. Relative error of $(\|\mathcal{X}^{t+1} - \mathcal{X}^t\|_F / \|\mathcal{X}^t\|_F)$ versus increasing iterations (a) for the *WDC* dataset recovered via LFCTNN-FL and (b) for the *cave* dataset recovered by LFCTNN-FC.

TABLE III
COMPUTATIONAL TIME (IN SECONDS) OF ALL TEST METHODS

	PU	WDC	*balloons*	*cd*
CSTF	143	152	153	154
LTRA	53	58	77	78
LTMR	50	51	187	188
CTRF	148	144	124	126
Exinl	28	28	115	119
GTNN	108	112	347	337
LFCTNN-FC	141	153	275	257
LFCTNN-FL	192	209	332	330



(a) (b)

Fig. 14. Ablation experiments on (a) WDC dataset using LFCTNN-FC and (b) *cd* dataset using LFCTNN-FL.

matrix, leading to a significant increase in runtime. Overall, our method demonstrates mixed performance compared to GTNN in terms of computational time, suggesting that the transformation domain has a limited effect on efficiency.

3) Roles of Full-Scale and Log-Sum-Based Scheme: To demonstrate the respective effectiveness of FCTNN and the log-sum-based scheme, ablation experiments are designed to investigate the roles of different components. We evaluate the contribution of each term to the final performance by removing them individually. Quantitative results of LFCTNN-FC on the WDC dataset and LFCTNN-FL on the *cd* dataset are presented in the form of radar charts in Fig. 14, where "w/o Log" indicates the absence of the logarithmic function. In addition, "$\lambda_1 = 0$" signifies that other terms are active, but the first mode of CTSVD is deactivated. Among these comparisons, LFCTNN-FC and LFCTNN-FL behave consistently better than their counterparts that arbitrarily remove certain priors. Furthermore, the application of CTNN along the first and third modes leads to a marked improvement in fusion accuracy, whereas the enhancement provided by CTNN along the second mode is relatively limited. This is likely because dimensionality reduction is initially performed on the second dimension. In addition, both λ_1 and λ_2 correspond to nonlocal spatial similarity, which displays a high degree of coherence. Notably, when the logarithmic function is omitted, a rapid performance decline is observed, highlighting the efficacy of the nonconvex strategy in capturing the essential characteristics of singular values.

V. CONCLUSION

Focusing on the challenge of HSISR, this work introduces two novel methods, LFCTNN-FC and LFCTNN-FL, utilizing a cascade-transform-based TSVD and TNN. Specifically, we propose a dual-enhanced CTSVD that is applicable to all modes, where DFT is combined with DCT and a self-learning transform. In addition, we derive CTNN as a convex relaxation of the tensor CT rank to mitigate computational limitations and develop a lossless transformation that induces tensors with lower average rank. Building on this, we present an unsupervised HSISR method by applying the newly developed CTNN to grouped patches, effectively exploiting the nonlocal spatial–spectral similarities in hyperspectral images. Extensive experimental results on both simulated and real-world datasets demonstrate the superior fusion quality achieved by the proposed methods.

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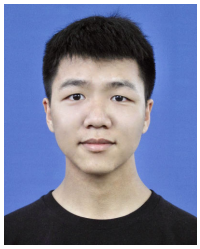
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